

Program 1:

Step1: Create a directory named program-1

mkdir program-1

cd program-1

Step 2: create the Dockerfile and add the similar contents

nano Dockerfile

```
GNU nano 7.2 Dockerfile *
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

Step 3: create a file -> [app.py](#) and add similar contents

nano app.py

```
from flask import Flask

app = Flask(__name__)
@app.route("/")
def home():
    return "Hello from Simple Flask Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 3: make a file -> requirements.txt
nano requirements.txt

Flask==2.3.3

Step 4: Build and run the docker command

docker build -t program-1 .

```
[+] Building 273.8s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                 0.0s
=> => transferring dockerfile: 355B                                0.0s
=> [internal] load metadata for docker.io/library/python:latest    7.2s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2 262.8s
=> => resolve docker.io/library/python:latest@sha256:1ad1a43b5e247 0.0s
=> => sha256:1ad1a43b5e2478e62056bbc28028afd8581 10.95kB / 10.95kB 0.0s
=> => sha256:0134b02fcae7fd813645ca13d87412024d3c4 6.48kB / 6.48kB 0.0s
=> => sha256:c4a87c25190f591b29fb62b4ee424ea9907b4 2.32kB / 2.32kB 0.0s
=> => sha256:13cc39f8244ac66bf1dd9149e1da421ab 49.29MB / 49.29MB 125.4s
=> => sha256:e3143549f2b8b3ad8d79efdc47824641c6 25.62MB / 25.62MB 71.3s
=> => sha256:72e8e93b0d018b135d833207c6feaba20 67.78MB / 67.78MB 112.6s
=> => sha256:23d55a674b2e3641a37b7f2053f00af 235.94MB / 235.94MB 257.9s
=> => sha256:8f5fb10be9d4525dc6941af4e2a587d1d3e 6.09MB / 6.09MB 124.6s
=> => sha256:2fbf02c04055ee28543076a502ce51fa4 29.23MB / 29.23MB 153.2s
=> => sha256:149075fccccd1b52406789bff7063813f1c14baf 248B / 248B 126.3s
=> => extracting sha256:13cc39f8244ac66bf1dd9149e1da421ab1bbc80d61 0.8s
=> => extracting sha256:e3143549f2b8b3ad8d79efdc47824641c6771796b3 0.3s
=> => extracting sha256:72e8e93b0d018b135d833207c6feaba205653ac52e 1.0s
=> => extracting sha256:23d55a674b2e3641a37b7f2053f00af6fe4f2a8957 4.0s
=> => extracting sha256:8f5fb10be9d4525dc6941af4e2a587d1d3edbbb744 0.1s
=> => extracting sha256:2fbf02c04055ee28543076a502ce51fa41895e386e 0.4s
=> => extracting sha256:149075fccccd1b52406789bff7063813f1c14baf10b 0.0s
=> [internal] load build context                                    0.0s
=> => transferring context: 638B                                       0.0s
=> [2/5] WORKDIR /app                                              0.2s
=> [3/5] COPY requirements.txt .                                    0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt        3.4s
=> [5/5] COPY . .                                                  0.0s
=> exporting to image                                              0.1s
```

docker run -p 5000:5000 program-1

```
harsha@HARSHA:~/program3$ sudo docker run -p 5000:5000 program3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. U
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
172.18.0.1 - - [07/Nov/2025 04:49:17] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [07/Nov/2025 04:49:17] "GET /favicon.ico HTTP/1.1" 404 -
q
□
```